



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

BLAZOR PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 Q&A on component lifecycle, data binding, JS interop, and state

TOTAL: 10 QUESTIONS

Q1 What is Blazor and what are the differences between its hosting models? **Seat Model**

Q6 How does Blazor routing work? **Observability**

Q2 What is the Blazor component lifecycle? **Architecture**

Q7 How do you validate a form in Blazor? **Lifecycle**

Q3 How does two-way data binding work with @bind in Blazor? **Blazor**

Q8 How do you manage shared state in a Blazor application? **Blazor**

Q4 How does Blazor JavaScript interop work? **Hardening**

Q9 What are RenderFragment and templated components in Blazor? **Testing**

Q5 What are cascading parameters in Blazor? **SSO / IdP**

Q10 How do you make HTTP requests in Blazor WebAssembly? **Testing**



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Blazor Server runs C# on the server

Mitigation

Blazor Server runs C# on the server and uses a SignalR WebSocket to push UI diffs to the browser. low initial load, requires persistent connection. Blazor WebAssembly runs the .NET runtime in the browser as a WASM binary offline-capable, larger initial download.

A6 @page '/users/{Id:int}' at the top of a

Observability

@page '/users/{Id:int}' at the top of a component registers the component as a route. `NavigationManager.NavigateTo('/home')` navigates programmatically. The Router component in App.razor matches the URL to a component and renders it in the Found template.

A2 OnInitialized(Async) runs once on mount.

Primitives

`OnInitialized(Async)` runs once on mount. `OnParametersSet(Async)` runs when parameters change. `ShouldRender` controls re-render. `OnAfterRender(Async)(firstRender)` fires after the DOM is updated. `StateHasChanged()` triggers a re-render manually from async callbacks.

A7 Wrap inputs in <EditForm Model='person'

Automation

Wrap inputs in `<EditForm Model='person' OnValidSubmit='HandleSubmit'>`. Add `<DataAnnotationsValidator />` to wire DataAnnotations rules and `<ValidationSummary />` to display all errors. Per-field errors appear with `<ValidationMessage For='() => person.Email />`.

A3 @bind='property' is syntactic sugar for value='@property'

Rendering

@bind='property' is syntactic sugar for value='@property' plus `oninput/onChange='@(e => property = e.Value)'`. @bind:event='oninput' changes the trigger event. @bind works on input, select, and textarea elements, and on child component parameters.

A8 Register a Scoped service (per-circuit in Blazor

Governance

Register a Scoped service (per-circuit in Blazor Server, per-session in WASM) that holds state. Inject it into components. Call `StateHasChanged()` in components after the service mutates state. Use events or Action callbacks on the service to notify subscribing components.

A4 Inject IJSRuntime. Call await

Gotchas

Inject `IJSRuntime`. Call `await JS.InvokeVoidAsync('window.alert', 'hello')` to call JS from C#. In JS, `DotNet.invokeMethodAsync('Assembly', 'StaticMethod', args)` calls a `[JSInvokable]` static C# method. For instance methods, use `DotNetObjectReference.Create(this)`.

A9 RenderFragment is a delegate representing a chunk

Assessment

`RenderFragment` is a delegate representing a chunk of Blazor markup. Expose a `[Parameter] RenderFragment ChildContent` and render it with `@ChildContent` in the template. Templated components accept `RenderFragment<T>` to give the caller control over rendering each item.

A5 <CascadingValue Value='theme'> in an ancestor propagates

Propagation

`<CascadingValue Value='theme'>` in an ancestor propagates theme to all descendant components that declare `[CascadingParameter] ThemeConfig Theme`. This avoids prop-drilling for cross-cutting concerns like themes, auth state, and localization.

A10 Register a typed HttpClient with

Optimization

Register a typed `HttpClient` with `builder.Services.AddHttpClient<WeatherClient>(() => c.BaseAddress = new Uri(builder.HostEnvironment.BaseAddress))`. The browser's fetch API handles the request under the hood. Use `GetFromJsonAsync<T>()` and `PostAsJsonAsync()` from `System.Net.Http.Json`.